



Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models

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Presenter: Emmanuel.

Good morning everyone. Today we present our work on Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models, focusing on extracting interpretable features from chest X-ray embeddings.



Introduction

Why Are Medical VLMs Opaque?

- Medical **Vision-Language Models** (VLMs) encode images and text in a shared representation with strong clinical performance
- But their latent representations are **high-dimensional** and **polysemantic**
 - Each dimension mixes multiple clinical meanings
- This limits **interpretability** and **trust** in safety-critical applications
- **Concept-based explanations** aim to map representations onto recognizable anatomy and findings

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Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models

└ Introduction

Medical VLMs like BiomedCLIP encode images and text in a shared space with excellent performance. However, their latent representations are polysemantic: each dimension mixes multiple clinical meanings, making it hard to understand what the model has learned. This limits interpretability and trustworthiness, which are critical in safety-sensitive applications like medicine. Concept-based explanations try to solve this by mapping these opaque representations onto recognizable clinical concepts.

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Introduction

Our Approach

- Following **MedConcept**: extract SAE features **without diagnostic supervision**, name them, and evaluate against radiology reports
- **Adaptation** from the original pipeline:
 - 3D abdominal CT → **2D chest radiographs**
 - Backbone: **BiomedCLIP** (frozen ViT-B/16)
 - New vocabulary and activation selection
- **Objective**: Assess what the evidence supports about interpretability — *not* a diagnostic classifier

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Introduction

Following MedConcept, we use a Sparse Autoencoder to decompose embeddings into sparse latent features without disease labels, and validate them against actual radiology reports. We adapted the original pipeline by swapping the backbone to BiomedCLIP and building a new dedicated concept vocabulary for 2D chest X-rays. Important: our goal is not to build a classifier, but to assess what the evidence actually tells us about interpretability.

Introduction

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Introduction

Key Contributions & Findings

Contributions:

1. An implemented pipeline with **phrase-level semantic grounding**
2. Analysis of concept groups at **multiple clustering resolutions**
3. An independent **multi-evaluator audit** of the LLM judge

Key Findings:

- Report-based alignment is **low on average (7.21%)** but **highly uneven** across clinical subdomains (up to 49.23% in specific clusters).
- The choice of judge is a major source of disagreement: **MedGemma agrees with human consensus only 56%** of the time vs. **68–76% for commercial LLMs**.

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Introduction

Our three main contributions are: a fully implemented pipeline with phrase-level grounding, an analysis of concept groups at different clustering resolutions, and an independent audit of the LLM judge. Our key findings show that while the average report-based alignment is low, around 7.2 percent, it's highly uneven across subdomains. Crucially, we found that MedGemma agreed with human consensus only 56 percent of the time, compared to 68 to 76 percent for commercial models, highlighting the judge's reliability as a major variable.

Introduction Key Contributions & Findings

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Related Work

From Predefined Concepts to Unsupervised Discovery

- **Supervised concept methods** (TCAV, Concept Bottleneck Models): semantic interfaces, but require **predefined annotations**
- **Sparse dictionary learning**: decomposes polysemantic activations into monosemantic features *without* predefined concepts
- Recent approaches go further:
 - First *discover* features, then *name* them — decoupling representation from labeling
 - Decompose VLM embeddings directly into sparse combinations of existing textual concepts
- In the **medical domain**: recent works dissect neuron activations in diagnostic models to extract concept-level explanations from opaque networks

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Related Work

Traditional methods like TCAV and Concept Bottleneck Models need predefined annotations. Sparse dictionary learning removes this by extracting monosemantic features unsupervisedly. Recent approaches decouple discovery from naming, or decompose VLM embeddings directly into textual concepts. In the medical domain, several works extract interpretable concepts from opaque diagnostic networks, which motivates our approach.

Related Work
From Predefined Concepts to Unsupervised Discovery

- **Supervised concept methods** (TCAV, Concept Bottleneck Model): semantic interfaces, but require **predefined annotations**
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Research Gaps

Four Gaps Guide Our Analysis

1. Reconstruction $\neq$ Correct Naming

Vocabulary coverage and image-text geometry constrain naming

2. Dependence on External Resources

Ontology, generative phrasing, LLM judges — errors can be mistaken for representation properties

3. Reports as Selective Evidence

Omission $\neq$ absence — judges can confuse the two

4. Independent Labels Omit Relationships

Embedding similarity $\neq$ anatomical hierarchy

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Research Gaps

Four gaps guide our analysis. First: good reconstruction does not mean correct naming. The vocabulary and the embedding geometry constrain what labels are even possible. Second: the pipeline depends on external resources at every stage. Errors in these tools can look like properties of the representation itself. Third: reports are partial. Radiologists omit normal anatomy, and the judge can confuse omission with contradiction. Fourth: assigning labels independently ignores relationships between concepts.

Research Gaps

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4. **Independent Labels Omit Relationships**
Embedding similarity $\neq$ anatomical hierarchy



Research Gaps

Our Response

- **Hierarchical clustering** at multiple resolutions to examine concept groups
- **Human + multi-model audit** on 25 stratified pairs to assess **evaluator reliability**
 - 3 human annotators + 3 commercial LLMs
- **Label permutation** to mitigate position bias in the LLM judge

→ *Goal: assess interpretability in a scalable and reproducible way*

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→ Goal: assess interpretability in a scalable and reproducible way

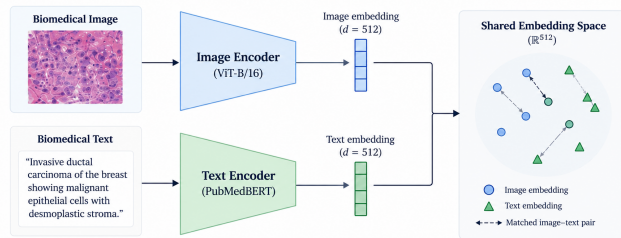
To address these gaps: we use hierarchical clustering to examine concept groups, a blinded audit with three annotators and three commercial LLMs to measure evaluator reliability, and label permutation to mitigate judge bias. The broader goal is assessing interpretability in a way that is scalable and reproducible.



Methodology

Datasets and Feature Extraction

- **NIH Chest X-ray 10k:** 7,500 SAE training / 2,500 validation images; no text or labels.
- **Open-I:** 3,666 frontal image-report pairs form the semantic evaluation. We combine Findings and Impression into reference reports to check extracted concepts.
- **Features:** Frozen BiomedCLIP ViT-B/16 image encoder; **512-dimensional embeddings**.



BiomedCLIP overview

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Methodology

- **Two datasets, different roles**
- **NIH Chest X-ray:** To train the SAE. Train only on image embeddings, without disease labels or reports.
- **Open-I:** Keep frontal images. Combine **Findings** (radiological observations) and **Impression** (diagnostic interpretation) into a reference report to check whether the extracted concepts were supported.
- **Frozen BiomedCLIP image encoder:** Convert each X-ray into a **512-dimensional embedding**, the input to the SAE.

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Methodology

Concept Dictionary Construction

- **Concept Selection:** 100 parent concepts from cleaned and filtered Open-I Problems metadata, covering anatomy and pathological findings.
- **Expansion:** UMLS provides Concept Unique Identifiers (CUIs) and synonyms from SNOMED CT and MeSH for matched concepts.
- **Phrase Generation:** Gemini 3.1 Flash-Lite generates 5 radiology-style descriptions per concept.
- **Encoding:** BiomedCLIP's PubMedBERT text encoder maps all 998 dictionary entries to L2-normalized, 512-dimensional embeddings in the same space as the images.

$$T \in \mathbb{R}^{998 \times 512}$$

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Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models

Methodology

- **To name the features:** learned by the SAE, we build a **dictionary of medical concepts**.
- **100 parent concepts:** From Open-I **Problems** metadata, filter devices, technical terms. Retain **anatomical structures and pathological findings**.
- **UMLS expansion:** Retrieve CUIs and synonyms from medical vocabularies. Keep the original term when no match exists.
- **Gemini 3.1 Flash-Lite:** Generate **five short radiology phrases per concept** in language of radiology reports, describing consistent visible features.
- **PubMedBERT text encoder:** Encode 498 terms/synonyms and 500 generated phrases. Matrix T contains **998 embeddings of 512 dimensions**, in the same space as the images.

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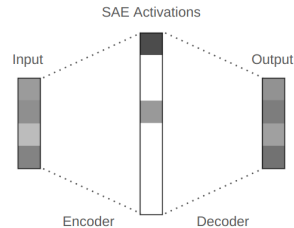
Methodology

Sparse Autoencoder Architecture and Training

- **Architecture:** 512-dimensional input, 1,024 hidden units (overcomplete factor of 2), ReLU activations z .
- **Objective:** Reconstruction error plus an L_1 penalty on activations; λ controls the sparsity penalty.

$$\mathcal{L} = \frac{1}{512B} \sum_{i=1}^B \|x_i - \hat{x}_i\|_2^2 + \frac{\lambda}{B} \sum_{i=1}^B \|z_i\|_1$$

- **Optimization:** Adam, batch size $B = 256$. Validation loss guides early stopping and model selection.



Illustrative SAE architecture

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- **Optimization:** Adam, batch size $B = 256$. Validation loss guides early stopping and model selection.



- **Goal:** Represent each image embedding as a **sparse combination of learned features**.
- **Architecture:** 512 input dimensions, 1,024 hidden units and ReLU activations. Overcomplete factor of **two**, as illustrated on the right.
- **Reconstruction loss:** Preserve information from the input embedding.
- L_1 **penalty:** Encourage sparse activations. λ controls the penalty strength.
- **Training:** Adam, batch size 256. Use **validation loss** for early stopping and model selection.



Methodology

Concept Assignment and Inference

After SAE training, each decoder column represents a learned feature direction in the image embedding space.

- **Naming:** Compare normalized decoder columns $\tilde{W}_d$ with dictionary embeddings T . Assign the entry with the highest cosine similarity to each feature.

$$S = T\tilde{W}_d \in \mathbb{R}^{998 \times 1024}$$

- **Inference:** Rank positive SAE activations for each Open-I image. Select features from largest to smallest until cumulative activation $> 90\%$ of the total positive activation.
- **Output:** Collect the selected features' labels and remove duplicate phrases to obtain the predicted concept set.

Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models

Methodology

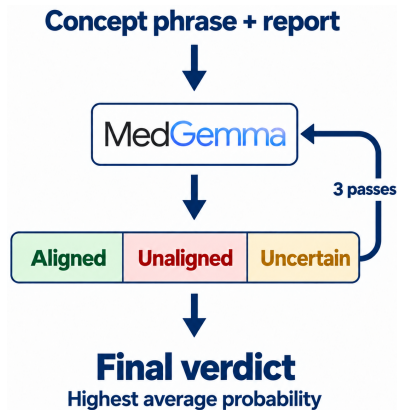
- After training, each **Decoder columns** represents a **learned feature direction** in the image embedding space.
- To name **Features**, compare normalized decoder columns with dictionary embeddings T using **cosine similarity**, producing matrix S .
- Assign each feature the **highest-similarity dictionary entry**.
- **Image inference:** Rank positive SAE activations from largest to smallest. Select features until cumulative activation **exceeds 90%** of the total positive activation.
- **Predicted concept set:** Collect labels and remove duplicates. The number of selected features varies across images.



Methodology

Semantic Evaluation and Validation

- **Evaluator:** Frozen MedGemma 1.5 4B; concept and report only (No image).
- **Verdicts:**
 - ✓ **Aligned:** Explicit support.
 - ✗ **Unaligned:** Explicit contradiction.
 - ? **Uncertain:** Unmentioned, ambiguous, or tentative.
- **Clustering:** Hierarchical parent-concept groups; To compare alignment across clusters.
- **Validation:** 25 concept-report pairs, 3 independent authors. Majority vote as reference for MedGemma and 3 commercial models.



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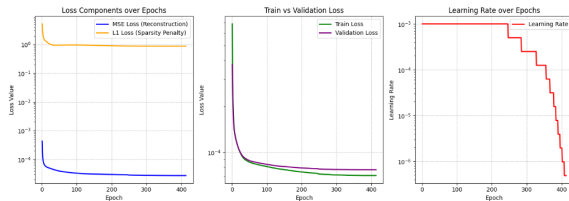


- To **evaluate** whether the extracted concepts are **supported** by the reference report. **Frozen MedGemma 1.5**, receives concept phrase and report, **without the image**.
- **Verdicts:** Aligned = explicit support. Unaligned = contradiction or negation. Uncertain = unmentioned or ambiguous.
- **Reduce letter bias:** Run three forward passes, rotating the A/B/C verdict mapping. Average verdict probabilities.
- **Hierarchical clustering:** Group related parent concepts and compare alignment across groups, reusing existing verdicts.
- **To check the judge's reliability:** Independently label **25 concept-report pairs**, blinded to MedGemma verdicts. Use author majority vote as the reference for MedGemma and **three commercial models**.



Experiments

Sparse Reconstruction and Visual Grounding



- **Reconstruction Fidelity:** Mean Squared Error (MSE) of 3.32×10^{-5} and mean cosine similarity of 0.9916 on 2,500 validation images.
- **High Sparsity:** Average L_0 sparsity of 44.0 active units out of 1024 per image ($\sim 4.3\%$ active neurons).
- **Dictionary Mapping:** Top 90% positive activations yield ~ 26 unique concept phrases per image.
- **Key Takeaway:** Near-lossless visual reconstruction is achieved under strict sparsity, but mathematical fidelity does not guarantee clinical monosemanticity.

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- **Key Takeaway:** Near-lossless visual reconstruction is achieved under strict sparsity, but mathematical fidelity does not guarantee clinical monosemanticity.

First of all: Let's look at how our Sparse Autoencoder reconstructs visual features.
As you see in the plots: MSE loss drops rapidly and stabilizes at 3.32×10^{-5} , with cosine similarity reaching 0.9916.
At the same time: High sparsity is maintained, activating only 44 out of 1024 units per image (4.3%).
However: High mathematical fidelity does not automatically mean features are clinically monosemantic.



Experiments

Report Alignment & Omission Bias

Images (N)	Phrases/img	Aligned (%)	Unaligned (%)	Uncertain (%)
10	26.20	7.25	23.17	69.57
50	26.22	8.40	22.88	68.72
100	26.23	7.64	23.37	68.99
500	25.99	7.21	22.49	70.30

- **Stable Verdict Distribution:** Across all sample sizes, $\sim 70\%$ of verdicts are *Uncertain*, $\sim 22.5\%$ *Unaligned*, and $\sim 7.2\%$ *Aligned*.
- **Reporting Omission Bias:** Radiologists document acute pathologies but omit normal anatomy, driving the dominant *Uncertain* rate.
- **Unaligned Rate (22.5%):** Captures both false-positive predictions and explicit clinical negations in reports.

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Experiments
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Moving to the second experiment: We tested how visual concepts match radiology reports using MedGemma as a judge.

Looking at the table: Across 500 images, results are very stable: 70% Uncertain, 22.5% Unaligned, and 7.2% Aligned.

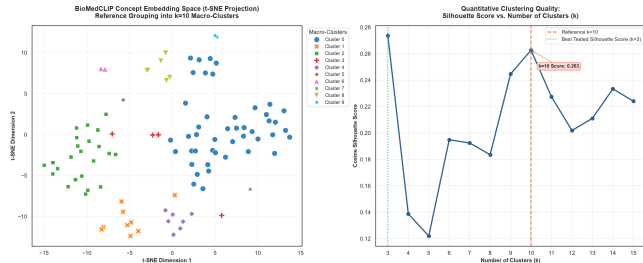
The main reason: The high Uncertain rate is driven by reporting omission bias (radiologists omit normal anatomy).

On the other hand: The 22.5% Unaligned rate represents false-positive predictions and explicit text negations.



Experiments

Hierarchical Clustering: Space & Quality



- **Concept Projection:** t-SNE visualization of the 100 parent concept embeddings grouped into $k = 10$ macro-clusters.
- **Silhouette Validation:** Quantitative analysis across $k \in [3, 15]$ shows a peak score of 0.274 at $k = 3$ and a near-peak score of 0.263 at reference $k = 10$.

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To analyze concept types: We clustered the 100 parent concept embeddings using cosine distance.

On the left plot: You can see the t-SNE visualization showing how concepts group into 10 macro-clusters.

On the right plot: Silhouette analysis confirms good separation, peaking at $k=3$ (0.274) and reaching 0.263 at $k=10$.

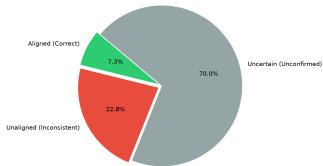
Therefore: We selected $k=10$ because it provides a detailed clinical grouping while keeping strong separation.



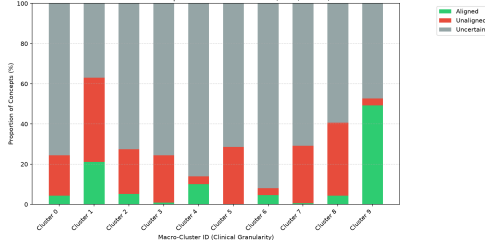
Experiments

Hierarchical Clustering: Subdomain Alignment

Overall Verdict Distribution Across Hierarchical Macro-Clusters (k=10, N=500)



Verdict Distribution per Semantic Macro-Cluster (k=10, N=500)

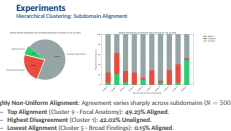


- **Highly Non-Uniform Alignment:** Agreement varies sharply across subdomains ($N = 500$).
 - **Top Alignment** (Cluster 9 - Focal Anatomy): **49.23% Aligned**.
 - **Highest Disagreement** (Cluster 1): **42.02% Unaligned**.
 - **Lowest Alignment** (Cluster 5 - Broad Findings): **0.15% Aligned**.

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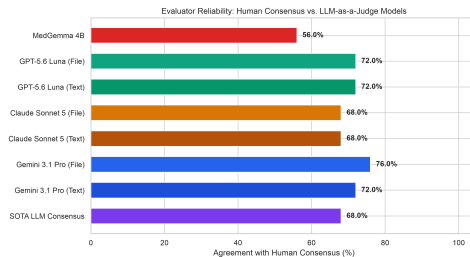


Examining alignment per cluster: We see that clinical agreement is highly non-uniform. **For instance:** Focal anatomical findings in Cluster 9 reach over 49% alignment (vs 7.2% global average). **In contrast:** Broad or mixed groups like Cluster 5 drop to almost 0% alignment. **This proves that:** The SAE captures specific anatomical features very well, and this pattern holds across all resolutions.



Experiments

Evaluator Audit & LLM Benchmark

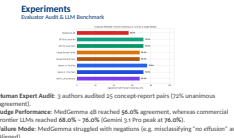


- **Human Expert Audit:** 3 authors audited 25 concept-report pairs (72% unanimous agreement).
- **Judge Performance:** MedGemma 4B reached **56.0%** agreement, whereas commercial frontier LLMs reached **68.0% – 76.0%** (Gemini 3.1 Pro peak at **76.0%**).
- **Failure Mode:** MedGemma struggled with negations (e.g. misclassifying “no effusion” as Aligned).

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Experiments



In our final experiment: We audited our automated LLM judge against a human consensus of 3 expert authors on 25 pairs.

As shown in the bar chart: MedGemma 4B agreed with human consensus only 56% of the time.

By comparison: Commercial frontier models like GPT-5.6, Claude 5, and Gemini 3.1 Pro achieved much higher agreement (up to 76%).

The main reason for this gap: MedGemma failed on medical negations (misclassifying “no pleural effusion” as Aligned).



Conclusions

Summary & Future Work

Summary of Key Findings:

1. **Mathematical vs Semantic Quality:** SAE achieves near-lossless visual reconstruction ($MSE = 3.32 \times 10^{-5}$, $L_0 \sim 4.3\%$), but text validation is bounded by reporting omission bias.
2. **Localized Interpretability:** Alignment is domain-specific; focal anatomical clusters reach up to **49.2% Aligned**.
3. **Evaluator Reliability:** Frontier LLMs significantly outperform MedGemma (76% vs 56% human agreement).

Future Directions:

- Refine concept dictionaries and validate feature assignments with clinical experts.
- Expand human audit datasets and incorporate spatial image grounding (heatmaps/bounding boxes).

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Conclusions

To summarize our main conclusions:

First: The SAE reconstructs visual features near-losslessly, but report validation is limited by omission bias.

Second: Feature interpretability is strongly localized: specific anatomical concepts align very well with clinical text.

Third: Automated evaluation requires strong LLMs, where frontier models clearly outperform smaller domain judges.

In the future: We plan to refine the concept dictionary with clinicians and test spatial grounding directly on image regions. Thank you!

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Thank you for listening!
Any questions?